

# TRANSFORMERS

## IN

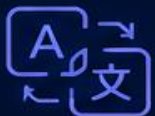
# NATURAL LANGUAGE PROCESSING

*From RNNs to Attention Mechanisms*

UNDERSTANDING LANGUAGE. POWERING INTELLIGENCE.



LANGUAGE  
UNDERSTANDING



MACHINE  
TRANSLATION



TEXT  
SUMMARIZATION



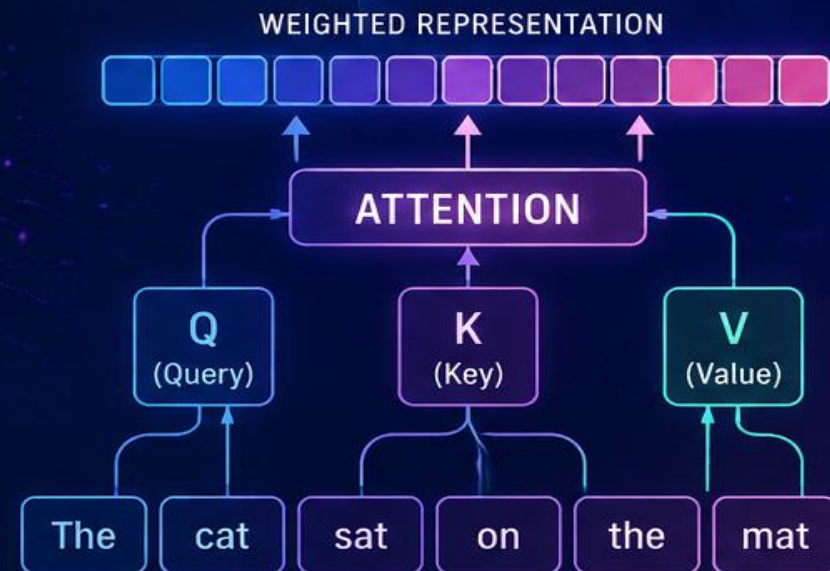
QUESTION  
ANSWERING



CHATBOTS



ATTENTION  
IS  
ALL YOU NEED





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## ■ Introduction to NLP :-

- NLP stands for **Natural Language Processing**.
- It is a branch of **Artificial Intelligence (AI)**.
- NLP enables computers to understand, interpret, and generate human language.
- Processes Text data, Speech data.

## ■ Goal of NLP :-

To enable effective communication between humans and machines using natural language.

## ■ Models used early days in NLP:-

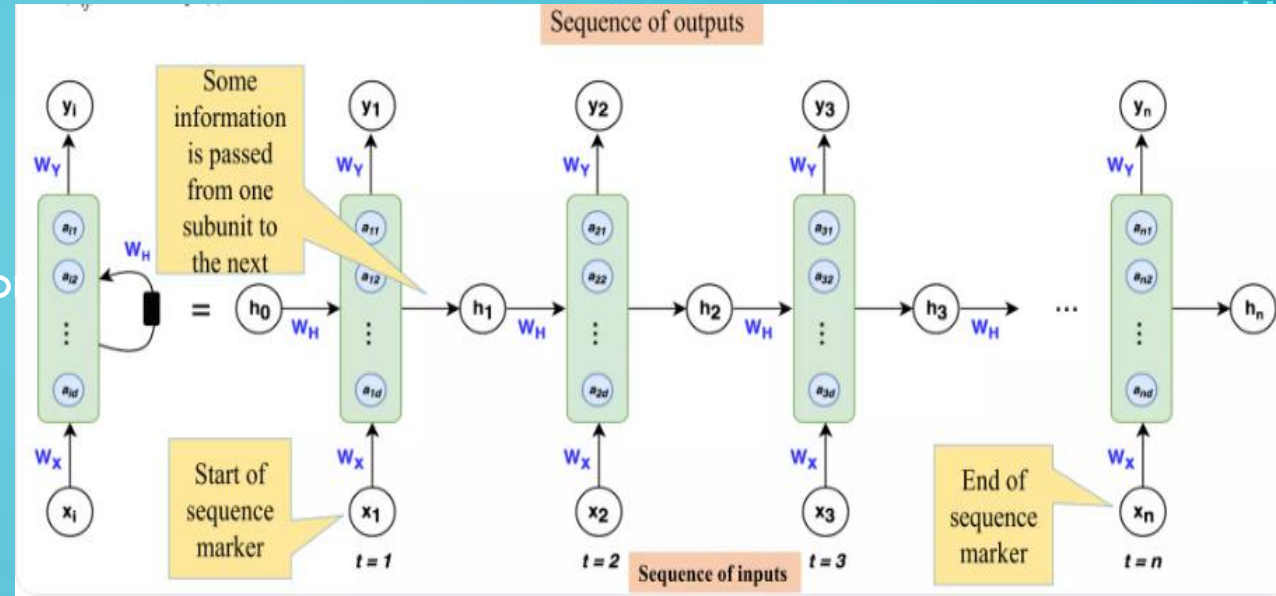
- Bag of Words (BoW)
- Hidden Markov Models (HMM)
- N-Gram Model
- TF-IDF (Term Frequency–Inverse Document Frequency)

## ■ Disadvantages:-

- Ignores word order.
- Cannot capture long dependencies.
- No contextual understanding.
- Limited sequence learning

## ■ Characteristics of RNN :

- Processes sequences sequentially.
- Uses hidden states to store previous information
- Shares weights across all time steps.
- Better than traditional NLP models.



## ■ Drawbacks of RNN :

1. RNNs struggle to remember information from very long sequences.

Q: The movie that I watched last week was amazing because \_\_\_\_ had great visuals.

2. Sequential Processing , No Parallelization .

RNNs process words one at a time. The next word cannot be processed until the previous step finishes. So the training becomes slow for long sequences and requires more training time.

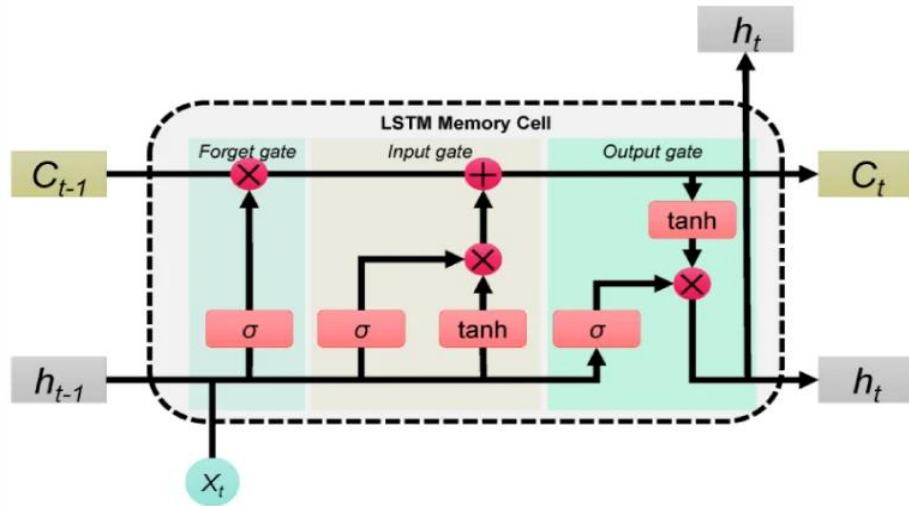
3. Sequential Processing , No Parallelization .

Example:

1. "I went to the **bank** to deposit money."
2. "I sat near the **bank** of the river."

RNNs failed to capture the full contextual information, especially in long sequences .

# ■ Long Short Term Memory(LSTM) :-



## 1. How LSTM Solves Long-Term Dependency Problem:

So LSTM introduces

1. Forget Gate
2. Input Gate
3. Output Gate

These gates control:

1. what to remember
2. what to forget
3. what to output

This helps important information flow across many time steps.

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

Where  $f_t$  represents forget unnecessary info, it store important info,  $C_t$  stores long-term memory.

## 2. How LSTM Partially Improves Context Understanding :

As LSTM can remember information longer, it captures context better than RNN.

But problems still exist. Very long sentences remain still difficult. Distant words in a long sentence still weaken over time.

In 2014, there was a paper named “Sequence to Sequence Learning with Neural Networks (2)” released by Ilya Sutskever and collaborators.

In this paper first introduce encoder decoder architecture for NLP.

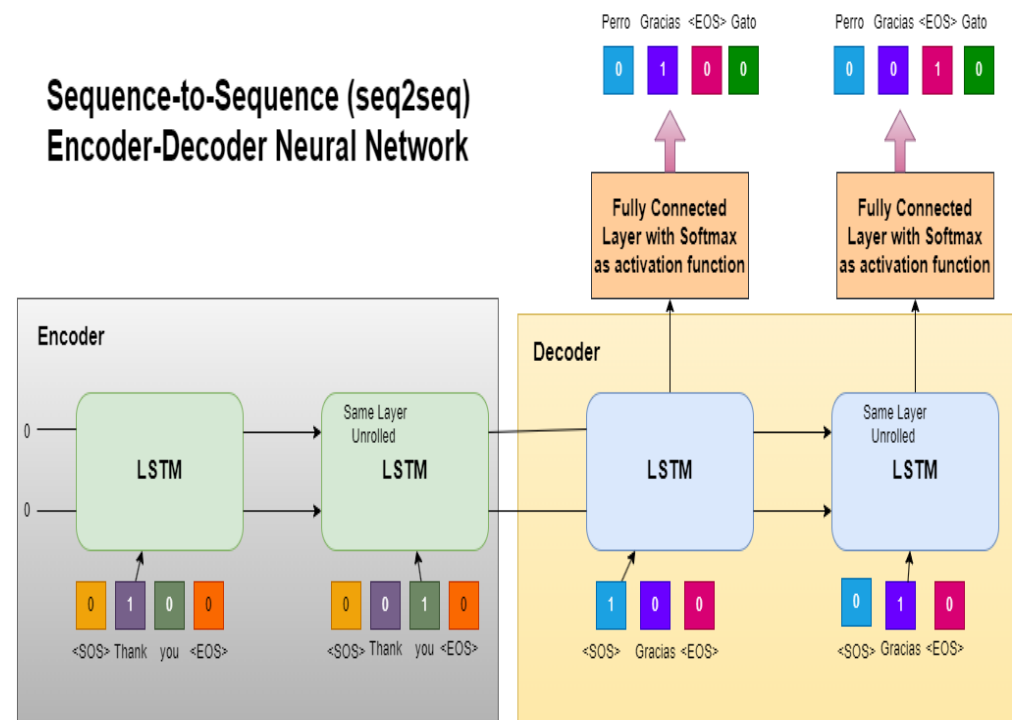
## ■ Advantages:

1. Handles Variable-Length Input and Output
2. Better Than Traditional RNN Models
3. The entire system is trained together. Encoder learns Representation and Decoder generates output.

## ■ Disadvantages:

1. Fixed-Length Context Vector Bottleneck.
2. Sequential Processing (Slow Training).
3. Limited Context Understanding.

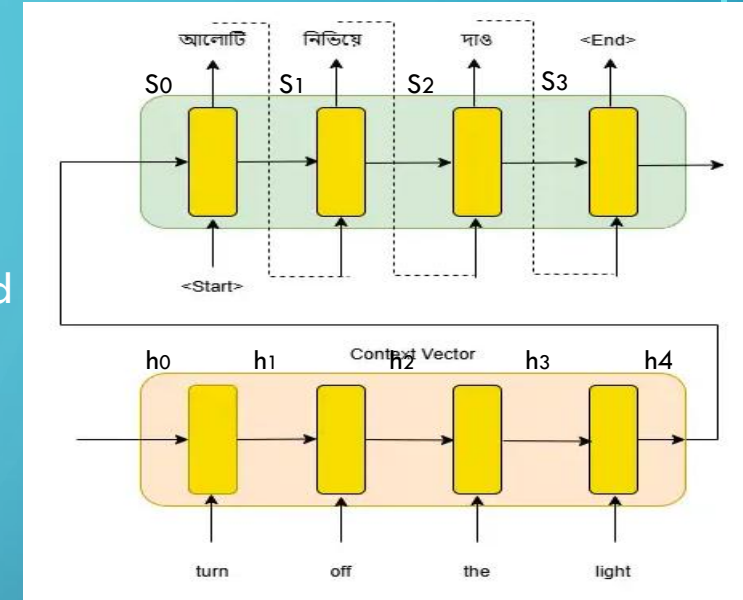
### Sequence-to-Sequence (seq2seq) Encoder-Decoder Neural Network



So, in 2014 there was released another paper named “Neural Machine Translation By Jointly Learning To Align And Translate (3)” by Dzmitry Bahdanau and collaborators .

In this paper first attention mechanism introduced.

The main idea of the paper is instead of compressing the entire input sentence into a single fixed vector, the decoder dynamically “attends” to relevant input words and different parts of the source sentence receive different attention weights during translation.

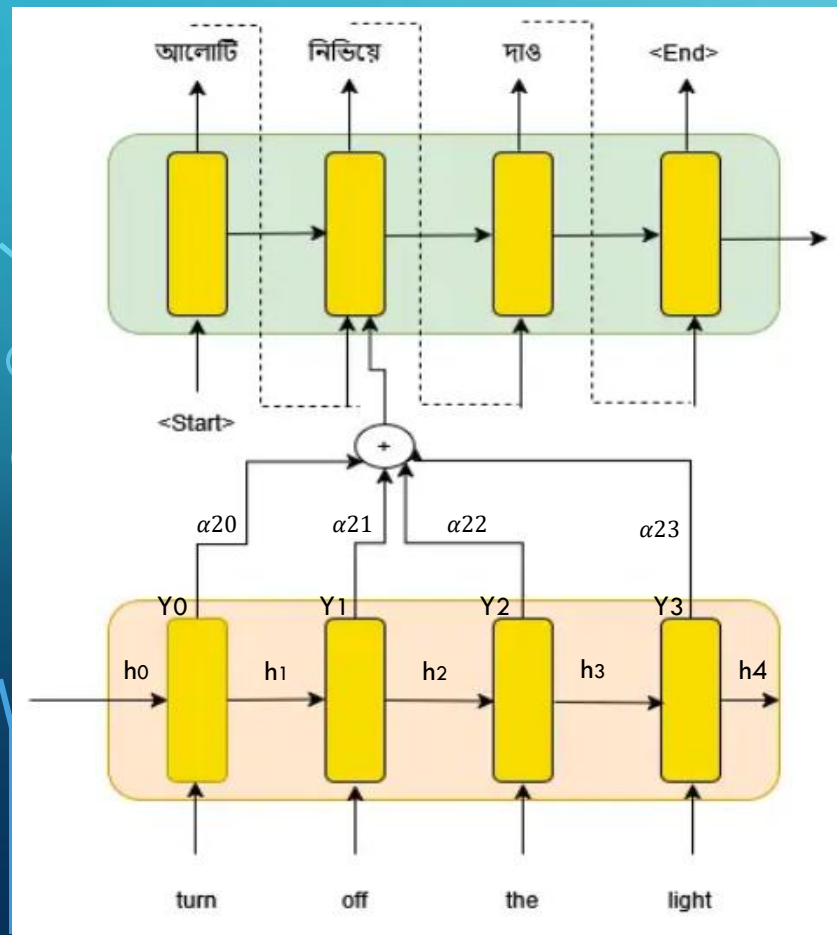


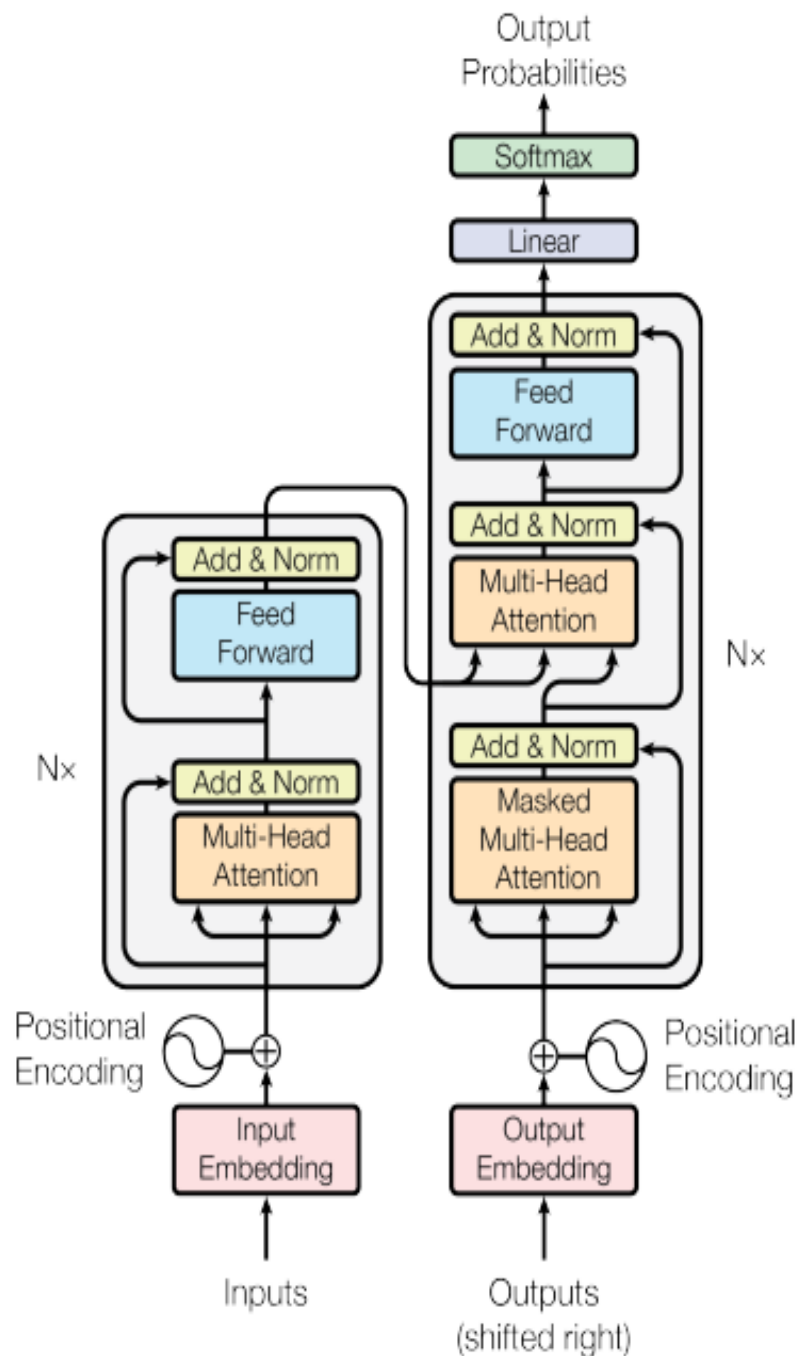
## ■ Advantages:

1. Solves Fixed Context Vector Bottleneck Problem
2. Improves the ability to capture the contextual information.
3. Slightly Better Handling of Long Sentences.

## ■ Disadvantages:

1. Still Sequential, results slow training.
2. Can't Apply Transfer Learning Concept.
3. Still difficult to Scale for Very Long Sequences or paragraphs.





This figure shows the original Transformer architecture proposed in the landmark paper “[\*Attention Is All You Need\* \(1\)](#)” by Ashish Vaswani and collaborators in 2017.

The architecture introduced the self-attention mechanism and replaced recurrent networks with a fully attention-based encoder–decoder framework, forming the foundation of modern large language models such as GPT and BERT.”

## ■ Why do we need attention?

The above RNN models does not effectively represent Contextual Word Representation.

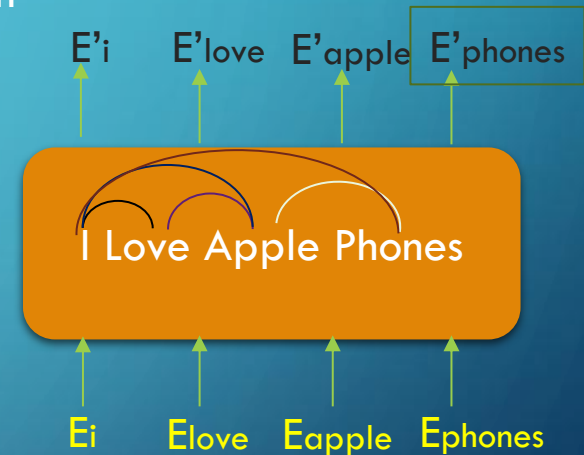
Eg:- I Love Apple Juice.

I Love Apple Phones.

we pass the word embedding of each word parallelly, and it finds the relationship between each word . Then it generate new word representation of each word.

## ■ How New Word Embedding Generated?

The context of the sentence is based on the words used in the sentence.

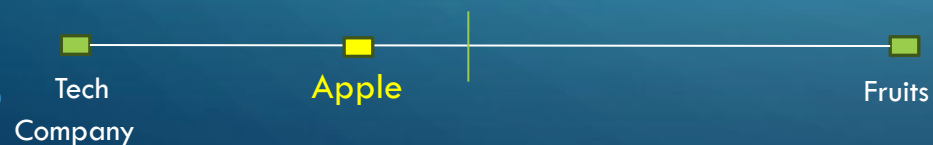


Love Apple Phones

$$E'_{love} = 0.8x E_{love} + 0.1x E_{apple} + 0.1x E_{phones}$$

$$E'_{apple} = 0.1x E_{love} + 0.5x E_{apple} + 0.4x E_{phones}$$

$$E'_{phones} = 0.1x E_{love} + 0.4x E_{apple} + 0.5x E_{phones}$$

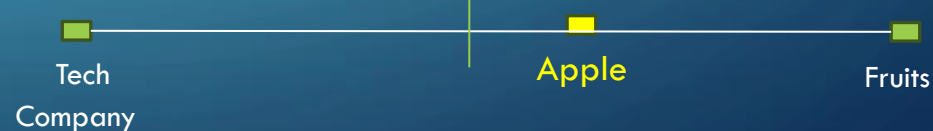


Love Apple Juice

$$E'_{love} = 0.8x E_{love} + 0.1x E_{apple} + 0.1x E_{juice}$$

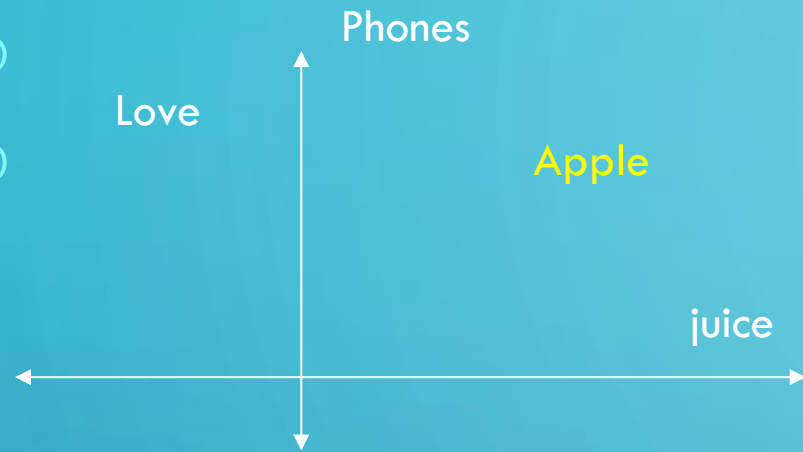
$$E'_{apple} = 0.1x E_{love} + 0.5x E_{apple} + 0.4x E_{juice}$$

$$E'_{juice} = 0.1x E_{love} + 0.4x E_{apple} + 0.5x E_{juice}$$



Here the no 0.8,0.1,0.5,0.4 represents the relationship between one word with other words.  
But how they are calculated ?

# How Self attention Works :-



We can assume the words as a multi dimensional vectors.

In mathematics, we find the distance/similarity between the vectors by dot(.) product of the vectors. So this 0.8,0.1,0.5 are the dot product of each word vector with other word vector.

But still there is a problem. The dot product can be negative, and the range of the dot product can be very large. But we want probabilities , so that we can effectively represents the relationship. So we use softmax function.

## Advantages:-

1. Solve the sequential training problem
2. Can capture general contextual information.

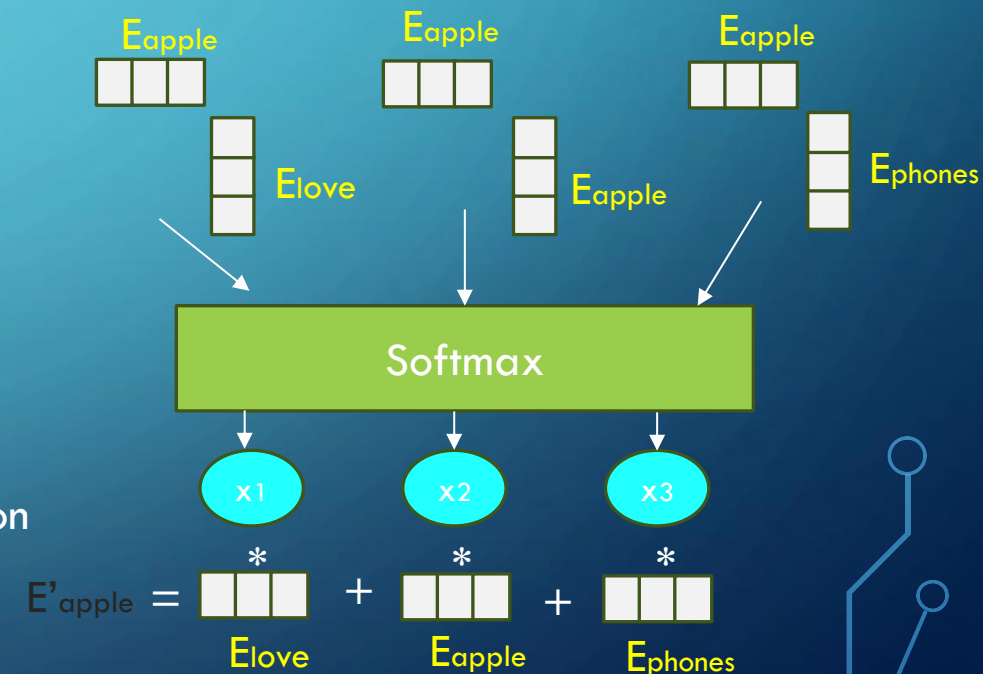
## Disadvantages:-

1. No Learnable Parameters :-

The above calculation is purely mathematics based calculation  
There is no learning parameters, so it can't learn from the input data.

2. Use of same vector:-

The same vector is worked as Key, Query And Value.  
So the performance is degraded.



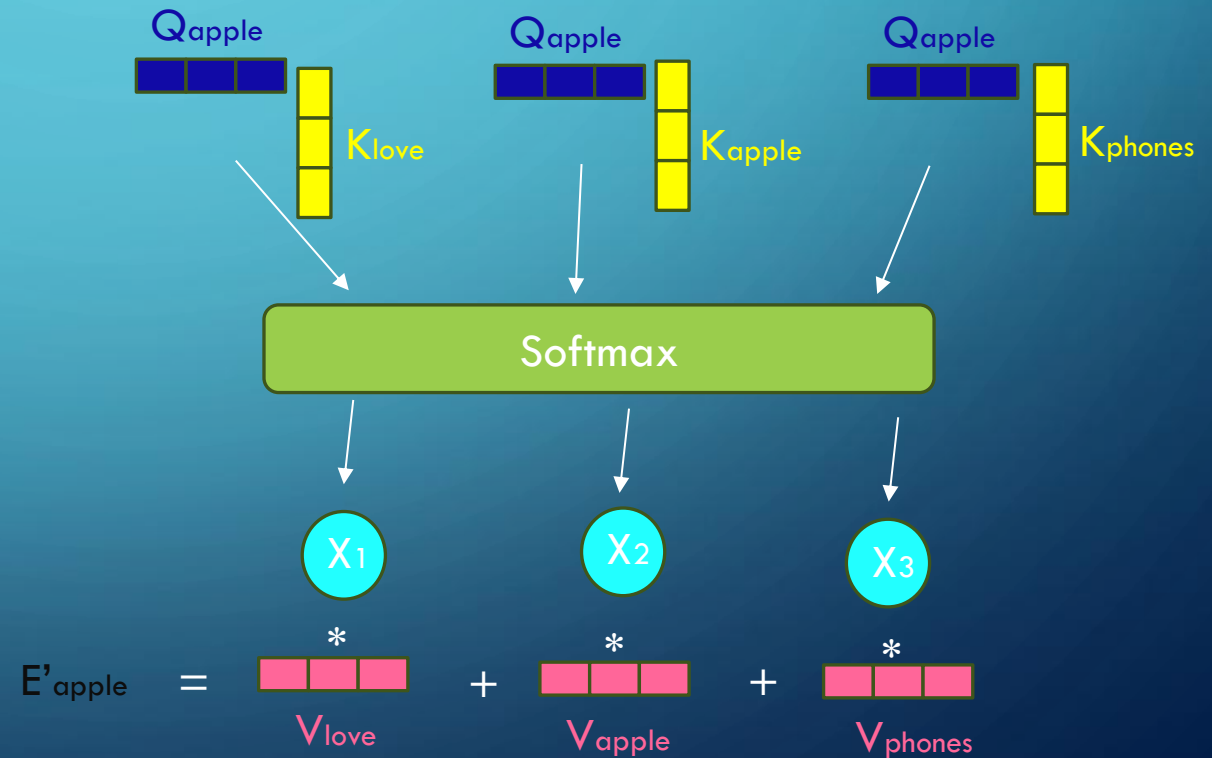
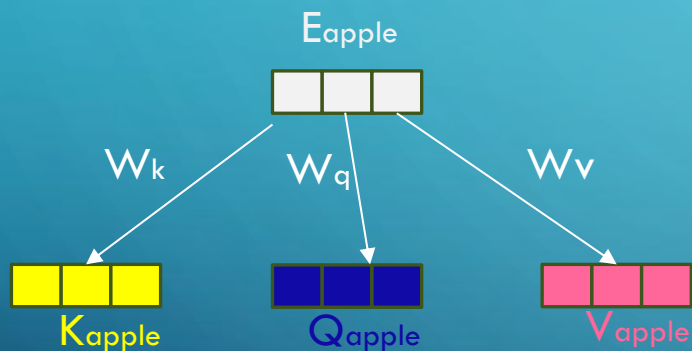
So we need to implement some weights and bias, whose value is changed during the training based on the input to implement task specific contextual information.

## But where we use these weights and bias ?

There are only two positions:- 1.Where we perform dot product of vectors.  
2.Where vector is multiplied with parameters.

## But How can we change the vectors ?

To change the vectors we perform linear Transformation, where we multiply the word vectors with the weighted vectors. Thus the above two problem solved.



This diagram represents a single headed attention. But in “Attention Is All You Need” paper they use multiheaded attention.

# ■ Multiheaded Self attention :-

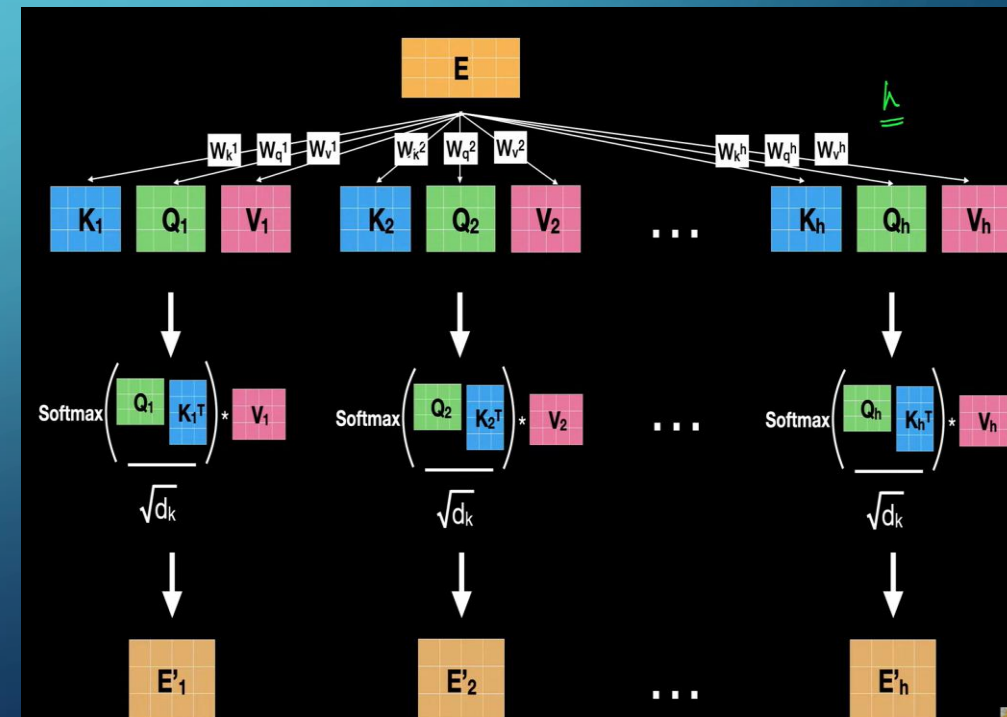
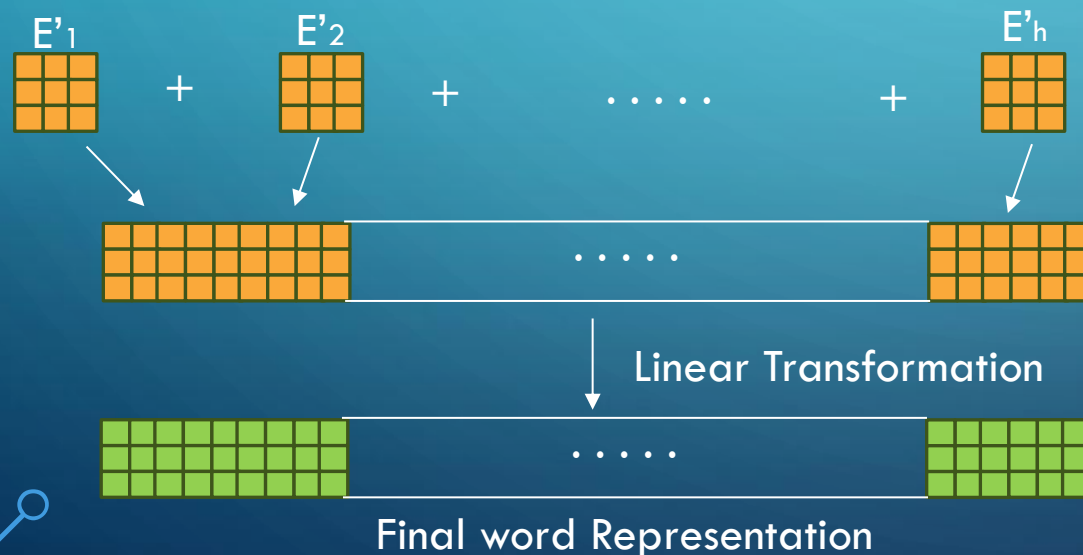
To understand the true meaning of language, we need to find complex relationship between words. It can be possible that sentence has ambiguous meaning and multiple interpretation.

Eg:- “He saw the man with the telescope”

The above sentence have two interpretation.

- 1.He saw the man who was holding the telescope.
- 2.He used a telescope to saw a man.

In multi headed attention we just simply use multiple self attention parallelly. So that different head of attention captures different sets of information.



## ■ Attention Formula:-

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Here  $d_k$  Represents dimension of  $k$  matrix.

we divide by  $\sqrt{d_k}$  (square root of the key dimension) to **prevent extremely large dot-product values**.

## ■ Positional Encoding:-

Transformers process words in parallel. They do not naturally understand word order.

Eg:- “Dog bites man” and “Man bites dog”.

For every word embedding:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

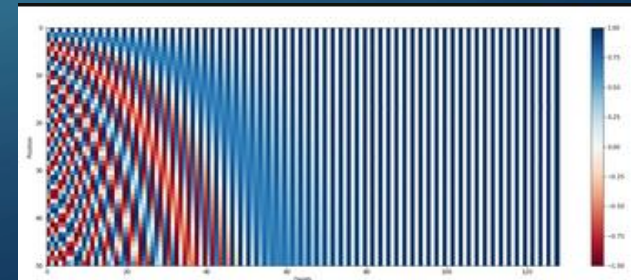
Input Representation = Word Embedding + Positional Encoding

The original transformer paper uses Sinusoidal Positional Encoding that uses sine and cosine functions.

Why Sine and Cosine ?

They provide:

1. Unique positional patterns
2. Smooth mathematical representation

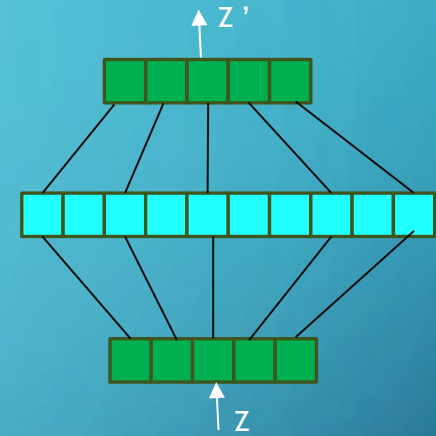


## ■ Feed Forward :-

**Feed Forward Neural Network (FFN)** processes the information obtained after the self-attention layer and helps the model learn complex patterns.

### Working Process:-

- Expands the feature dimension.
- Applies non-linearity (activation function)
- Compresses it back.



## ■ Masked Multi-headed Attention:-

Masked Multi-Head Attention is used in the **decoder** of a transformer to prevent the model from seeing future words during training.

When generating a sentence:

- the model should only use previous words
- not future words.

$$\text{Softmax} \left[ \frac{\text{Query Matrix} \times \text{Transposed Key Matrix}}{\sqrt{d}} \right] = \text{Unnormalized Attention Scores} \rightarrow \text{Masked Attention Scores}$$

|      |      |      |      |      |
|------|------|------|------|------|
| 1.1  | 2.3  | 5.3  | 2.1  | -1.2 |
| -0.1 | 0.5  | 1.3  | -0.1 | 6.1  |
| 2.3  | 0.2  | 3.3  | -1.0 | -0.4 |
| 0.3  | 1.2  | -0.3 | 5.0  | 1.4  |
| 5.1  | -2.9 | -1.1 | -4.2 | 0.4  |

|      |      |      |      |      |
|------|------|------|------|------|
| 1.1  | -inf | -inf | -inf | -inf |
| -0.1 | 0.5  | -inf | -inf | -inf |
| 2.3  | 0.2  | 3.3  | -inf | -inf |
| 0.3  | 1.2  | -0.3 | 5.0  | -inf |
| 5.1  | -2.9 | -1.1 | -4.2 | 0.4  |

|      |       |        |
|------|-------|--------|
| love | apple | phones |
| love | apple | phones |
| love | apple | phones |

|   |      |      |
|---|------|------|
| 0 | -inf | -inf |
| 0 | 0    | -inf |
| 0 | 0    | 0    |

|      |       |        |
|------|-------|--------|
| love | -inf  | -inf   |
| love | apple | -inf   |
| love | apple | phones |

Thus the multi headed attention solves all the previous problems.

## ■ **Advantages :-**

1. Captures Long-Range Dependencies.
2. Better Contextual Understanding.
3. Parallel Processing (Fast Training).
4. Can use as Transfer Learning.
5. Captures Long-Range Dependencies.

## ■ **Disadvantages :-**

1. High Computational Complexity.
2. Large Memory Consumption.
3. Requires Large Datasets.
4. Expensive Training.

## ■ Applications :-

1. Machine Translation
2. Text Summarization
3. Chatbots & Conversational AI
4. Vision Transformers (ViT)
5. Speech Processing

## ■ Popular Models :-

1. BERT
2. GPT
3. Claude
4. LLaMA

## ■ References:-

### 1. “Attention Is All You Need” .

Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

### 2. “Sequence to Sequence Learning with Neural Networks” .

Sutskever, Ilya, Oriol Vinyals, and Quoc V. Le. "Sequence to sequence learning with neural networks." *Advances in neural information processing systems* 27 (2014).

### 3. “Neural Machine Translation By Jointly Learning To Align And Translate” .

Bahdanau, Dzmitry, KyungHyun Cho, Yoshua Bengio, and Present By Jing Huang. "Neural Machine Translation By Jointly Learning To Align And Translate."

